

Project Proposal

Practical Machine Learning — Green Data Science 2025/2026

Project Title	Predicting Drinking Water Potability from Chemical Properties
Category	Tabular data, binary classification
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1. Problem Statement

Access to clean water is something most people take for granted, but in many parts of the world it remains a serious problem. Contaminated drinking water causes millions of deaths every year, and even in places with better infrastructure, water quality monitoring is often slow and expensive.

The idea for this project came from wanting to apply what we learned in class to something with real environmental relevance. The goal is to predict whether a water sample is safe to drink based on its chemical properties like pH, hardness, chloramines and turbidity. This is a binary classification problem: potable (1) or not potable (0).

If a model like this works well, it could potentially support automated water quality monitoring systems, reducing the need for constant manual lab testing. From a Green Data Science perspective, efficient water management is one of the most pressing sustainability challenges, so this felt like a natural fit for the course.

2. Challenges

A few things I expect to be tricky:

- **Missing data:** The dataset has quite a lot of missing values, especially in pH and sulfate. I will need to decide how to handle these carefully, probably using median imputation or KNN imputation, but only on the training data to avoid leakage.
- **Class imbalance:** The classes are somewhat imbalanced, there are more non-potable samples than potable ones. This means accuracy alone is not a good metric and I will need to focus on F1-score, precision, recall and ROC-AUC instead.
- **Feature correlation:** Some features are probably correlated, which could affect how the models interpret variable importance. I plan to look at this with SHAP.

- **Model selection:** Choosing the right model and hyperparameters without overfitting. I will use cross-validation with StratifiedKFold to make sure the evaluation is honest.

3. Dataset

I will use the Water Quality and Potability dataset from Kaggle (<https://www.kaggle.com/datasets/adityakadiwal/water-potability>). It has 3,276 water samples with 9 chemical features and a binary target variable.

The features are: pH, hardness, total dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. The dataset is small enough to include directly in the GitHub repository.

4. Proposed Method

I plan to follow the standard ML pipeline from the course:

- Start with some exploratory analysis to understand the data and spot any obvious issues
- Handle missing values with imputation (computed only on training data)
- Train/test split with stratification to preserve class balance — probably 80/20
- Standardize features using a sklearn Pipeline so that scaling is never applied to the test set before splitting
- Train and compare at least three models: Logistic Regression as a baseline, Random Forest, and XGBoost
- Tune hyperparameters with GridSearchCV using StratifiedKFold (5 folds)
- Analyse feature importance with permutation importance and SHAP

5. Evaluation

Because of the class imbalance, the main metrics will be F1-score and ROC-AUC. I will also report precision and recall to understand the trade-off between false positives and false negatives. In a water safety context, a false negative (saying water is safe when it is not) is much more dangerous than a false positive.

All evaluation will be done on the independent test set. I will compare results across models in a table and plot the ROC curve for the best performing one. I also want to include a SHAP summary plot to show which chemical properties matter most for the prediction.

6. Why This Is Relevant for Green Data Science

Water quality is directly connected to environmental sustainability and public health. This project applies the core ML concepts from the course like classification, cross-validation, ensemble methods, evaluation metrics and interpretability to a real world environmental problem. I think it strikes a good balance between being manageable as a solo project and being technically interesting enough to cover the main topics from class.